LFM2.5-Audio on AI Glasses: Feasibility Across Two Hexagon NPU Generations and a Native Phone Deployment

Ivan A. Novosad

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Abstract

We evaluate LFM2.5-Audio-1.5B as a speech-to-speech assistant for AI glasses. With no AR1-class target available, we profile fixed-shape components on two Qualcomm NPU generations (a QCS8550 proxy and an HTP v81 reference device — the target phone's own silicon generation), establish a full-model BF16 reference on an NVIDIA L4, run a Q4 quantization matrix on Apple M2 Metal, and ship the Q4 bundle as a native Android application on a Snapdragon SM8850 phone. Every neural partition compiles to a strict NPU context with zero CPU fallback on both generations, 1.46–1.88× faster on v81; the depth decoder returns exact tokens and the encoder passes its downstream acceptance test. The phone demo runs microphone to generated speech end to end with no host attached, at 0% WER on the bundled samples and first audio near 800 ms, and its 1.83 GiB residency is fully attributed — chiefly a 556 MiB CPU weight repack plus fixed compute buffers. Two measured results set the remaining agenda: FP16 detokenization is numerically unusable on both NPU generations, so it stays in FP32 on the host; and no tested CPU configuration clears the 1331 MiB memory gate, the 0.3 pp quality rule, and interactive latency at once. The NPU port is unblocked — toolchain gate passed, W8A16 pipeline validated end to end — and a sub-1B modular pipeline (Moonshine Base → Qwen3-0.6B → Pocket TTS) remains the deployment control.

1 Introduction

Speech-to-speech assistants on eyewear are constrained less by model quality than by what a wearable NPU will accept and what its RAM budget will hold. This report asks a narrow question about one candidate: can LFM2.5-Audio-1.5B [Liquid AI, 2026a] serve as the integrated baseline for a first device integration, and what evidence would be needed to call it glasses-ready?

The answer is a qualified yes on the research track and a not-yet on the product track. All fixed-shape neural partitions we tested compiled to strict NPU contexts with zero CPU fallback on the proxy device, and the official BF16 model completed ASR and a persistent two-turn speech interaction on an NVIDIA L4. The blockers are target-hardware verification, target-QNN resident memory, a production cache interface, and the output-audio path.

Hardware boundary. Every Qualcomm number in this report was measured on QCS8550 (Proxy), Android 12, Hexagon v73, through Qualcomm AI Hub [Qualcomm, 2026]. No AR1/AR1+/5100 target was available. Proxy figures are *placement evidence*, not AR1 performance, and this qualification applies to every Qualcomm table and figure below without further repetition.

Contributions.

- A strict-placement component profile of LFM2.5-Audio on a Hexagon NPU, with per-component latency, peak memory, and numerical acceptance (§3).
- Evidence that full NPU placement and numerical usability are independent properties, demonstrated by a fully-placed detokenizer that fails on real codes (§5).
- A static memory and KV-cache ledger showing why BF16 assets cannot be resident in a 2 GiB budget (§6).
- A native-CPU phone deployment whose residency is fully attributed (weight repacking, compute geometry, audio-length-scaled encoder activations), with a matched on-device 18-utterance diagnostic and a negative joint memory–quality–latency verdict (§8).
- An HTP v81 transfer matrix on the phone's silicon generation and a W8A16 quantization smoke, establishing the NPU port's toolchain path and its two open blockers (§9).
- A candidate comparison against Whisper, Moonshine, Zipformer, and Piper component cards, and the resulting baseline decision (§10).

2 Setup

2.1 Usage mode

The product mode is VAD-gated activation followed by a continuous conversational session. This avoids full-model inference while idle, but it does not bound active-session context: once a session opens, KV growth is unbounded until the session manager intervenes. An explicit reset, sum-

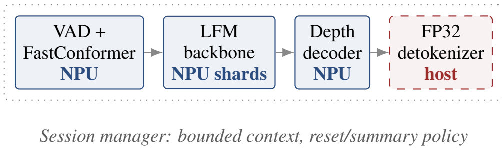

Figure 1: Recommended LFM partition for a first device integration. The host detokenizer is an architectural recommendation, not a measured on-glasses result.

Table 1: Scope of the experiments. The final row is the part that constrains every conclusion in this report.

Class	Contents
Local	Eight official-weight fixed-shape PT2 exports; repeated Q4_0/KV-cache matrix on Apple M2 Metal; static memory ledger.
Qualcomm proxy	Strict NPU placement, component-only latency and memory, local-vs-remote numerical comparison.
Reference GPU	Full BF16 ASR plus two-turn interleaved text/audio generation on one NVIDIA L4.
Phone (CPU)	Native Android ARM64 Q4 CPU deployment: batch latency, resident memory attribution, CPU core-seconds, thermal snapshots, matched 18-utterance diagnostic. Not an NPU path.
v81 reference (cloud)	Fixed-shape FP16 transfer and a W8A16 smoke on a Snapdragon 8 Elite Gen 5 QRD (HTP v81) via AI Hub — the phone's silicon generation, not the phone itself.
Quality	Downstream encoder sensitivity, exact depth tokens, generated-wave integrity, small matched ASR diagnostic.
Not measured	Full AR1 pipeline, host/NPU transfers, glasses RSS, power, VAD, microphone-to-speaker latency, an accepted quantized LFM QNN graph.

marization, or eviction policy is therefore a requirement, not a refinement. Figure 1 shows the partition we recommend for a first integration.

2.2 Evidence classes

Table 1 states what each class of measurement does and does not establish. The distinction matters throughout: local, proxy, and reference-GPU evidence are not interchangeable, and none of them is a glasses measurement.

3 Component placement on the proxy

Every tested strict graph placed fully on the Hexagon NPU (Table 2). Zero operators fell back to CPU in any partition. Latency spans an order of magnitude, from 1.880 ms for the four-frame detokenizer to 25.540 ms for the depth/RQ decoder, and peak memory is flat at 87–92 MiB across all eight graphs.

The two × rows are the finding. Placement and numerical acceptance are independent: the detokenizer compiles cleanly, runs fast, and produces audio that is unusable (§5). Section 9 repeats all eight graphs on the phone's HTP v81 generation; the pass/fail classes do not move.

Table 2: Strict-placement component results on QCS8550 (Proxy). NPU/CPU is the operator split; every graph placed with zero CPU fallback. Numerics: ✓ pass, × mismatch against the local golden. †Outside the deliberately strict 10^{-3} tensor gate (max abs 0.081) but passes the downstream acceptance test — 16/17 top-1 decisions preserved, golden in the top-5 throughout (§5) — which is the criterion this report uses.

Component	NPU/CPU	Latency	Peak	Num.
FastConformer + adapter	1215/0	4.786 ms	87.0 MiB	✓†
Conv layer prefill	36/0	2.693 ms	92.2 MiB	✓
Attention layer prefill	62/0	2.444 ms	91.5 MiB	✓
Conv cached decode	32/0	2.695 ms	91.9 MiB	✓
Attention cached decode	61/0	2.452 ms	87.0 MiB	✓
Depth/RQ decoder	2953/0	25.540 ms	92.0 MiB	✓
Detokenizer neural T=4	371/0	1.880 ms	88.0 MiB	×
Detokenizer neural T=8	371/0	2.577 ms	87.4 MiB	×

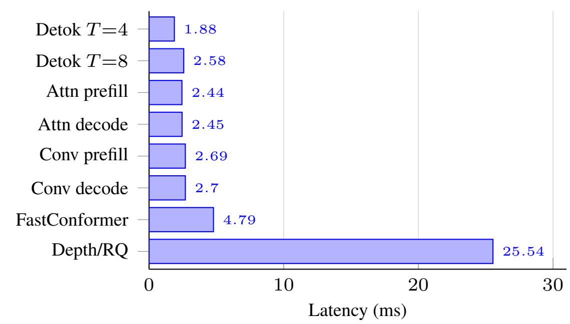

Figure 2: Per-component latency on the proxy. These values must not be summed: the components have different shapes, different invocation counts, different compiled-context overheads, and different transfer boundaries.

4 Full-model reference behavior

The official BF16 model completed a persistent two-turn speech interaction on a single NVIDIA L4 (Table 3). Turn 2 is uniformly faster than turn 1 because the session cache is warm: first text drops from 97.97 ms to 56.49 ms and first audio token from 339.67 ms to 225.87 ms.

In a separate sequential ASR pass the same checkpoint preprocessed in 0.559 s, produced its first token at 0.502 s, and generated 58 tokens in 1.593 s (36.40 tokens/s). Checkpoint load took 16.104 s.

The run used 3.034 GB steady and 6.216 GB peak CUDA allocation, with 6.388 GB reserved. These are reference-GPU allocator figures, not mobile RSS — §8 supplies the real-device number — and they are not a budget for the glasses. The official output detokenizer remained FP32 throughout.

Table 3: Full-model BF16 golden on one NVIDIA L4. Turn 2 benefits from a warm session cache.

Metric	Turn 1	Turn 2
First text	97.97 ms	56.49 ms
First audio token	339.67 ms	225.87 ms
Generated steps/s	15.74	16.35
Decoded waveform	11.84 s	8.96 s

Table 4: WER (%) on the same 18-utterance diagnostic; lower is better. Best per column in bold.

Model	Clean	White 10 dB	Compet. 5 dB
LFM2.5	**2.86**	**6.43**	55.71
Whisper Tiny	11.43	21.67	71.90
Whisper Base	8.81	20.71	55.00
Whisper Small	7.14	10.71	**31.43**
Moonshine Base	8.57	14.52	64.05

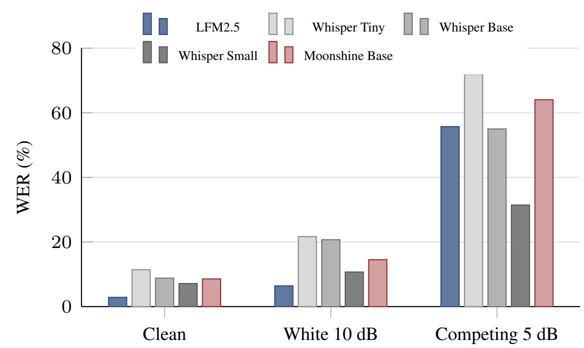

Figure 3: WER on the 18-utterance diagnostic. The clean-speech ranking does not survive competing speech.

5 Quality

5.1 Encoder fidelity

The encoder path passes its acceptance test. A frozen downstream LFM sensitivity check on the proxy features preserves 16/17 top-1 decisions — the single branch change is *seems* to *looks* — with the golden choice in the remote top-five at every step. Feature-space agreement is high (cosine 0.998424, NRMSE 0.06976, 100% nearest-frame temporal identity over ten captured frames) though outside the deliberately strict 10^{-3} tensor gate, and that combination is the point: acceptance is defined at task level, because the detokenizer below shows that tensor-adjacent metrics can pass placement and still produce catastrophic output. Every precision change is therefore re-checked on transcripts, tokens, and task behavior.

5.2 Detokenizer failure

The depth/RQ decoder returns exact audio code tokens. The FP16 NPU detokenizer, given those real generated codes, does not: waveform cosine 0.0052, NRMSE 1.010, and SI-SDR -45.74 dB over the first eight valid turn-1 frames. The graph still places 371/371 runtime layers on the NPU at 2.564 ms.

This is the single most consequential result in the report. A component can be fully accepted by the compiler, run at target speed, and be worthless. The first device integration should keep the official FP32 detokenizer outside the FP16 HTP path until a host implementation is measured and accepted.

5.3 Matched ASR diagnostic

We transcribed the same 18 LibriSpeech dummy utterances under three conditions: clean, deterministic 10 dB white noise, and 5 dB competing speech (Table 4, Figure 3). WER is normalized; lower is better. This is a smoke diagnostic on synthetic noise, not a product benchmark and not a glasses-microphone benchmark.

LFM is strongest on clean speech (2.86%) and on 10 dB white noise (6.43%). Under 5 dB competing speech the ordering inverts: Whisper Small leads at 31.43% while LFM degrades to 55.71%. Moonshine Base remains the footprint challenger, not a demonstrated multi-speaker quality winner.

On the diagnostic itself LFM's mean first-token latency was 101.7 ms clean, 55.3 ms under white noise, and 55.6 ms under competing speech, at real-time factors of 0.064–0.074 and 3.14 GB peak allocation. Those are L4 figures and are not comparable to the Apple MPS runtimes recorded for Moonshine, nor to any Qualcomm or phone latency in this report.

5.4 Generated-audio intelligibility

Retranscribing LFM's own generated audio with Whisper Small gives 0.00% WER for turn 1 and 21.43% WER for turn 2 against LFM's own generated text. The WAVs are finite, 24 kHz mono, and unclipped. This is an audio-text consistency proxy, not a naturalness or MOS score.

6 Memory and context budget

The pinned BF16 runtime checkpoint set occupies 3.390 GiB in static files. The complete local Q4 GGUF bundle occupies 1.001 GiB — a 3.39× smaller package, though across different formats and backends, so it is a package-to-package comparison and nothing stronger. Static file size is not process RSS. The consequence stands regardless: the BF16 assets cannot all be resident inside a 2 GiB RAM budget.

Active-session growth is dominated by the six attention-layer KV caches (Table 5); the fixed convolution cache is roughly 0.12 MiB at BF16 and never matters. VAD reduces idle work but does not touch this requirement — which is why the session policy of §2.1 is load-bearing.

Table 5: Ideal KV payload by context length. Tensor-payload estimates only; actual quantized runtime buffers differ because of scales and alignment.

Context positions	FP16 payload	INT8 payload
4,096	48 MiB	24 MiB
8,192	96 MiB	48 MiB
16,384	192 MiB	96 MiB
32,768	384 MiB	192 MiB

7 Local quantization matrix

The official Q4_0 GGUF bundle was run 40 times across two official audio samples with FP16, Q8_0, and Q4_0 KV caches plus a flash-attention control. All 40/40 normalized transcripts match exactly (Table 6). Q4_0 KV cuts cache memory from 48.0 MiB to 13.5 MiB with no transcript change on this sample.

This is an Apple M2 Metal sanity check. It is not Qualcomm QNN evidence, not AR1 evidence, and not a general quality benchmark. The runner reports Metal fallback hotspots for `CONV_2D_DW`, `ROLL`, and `UNARY` in the audio encoder; those are Metal hotspots, not QNN CPU-fallback evidence. No quantized LFM QNN graph has yet been profiled on QCS8550 or on the glasses.

8 Native phone deployment

8.1 What this measures

We packaged the same official Q4 GGUF bundle as a native Android application and profiled it on a Nubia NX809J (Snapdragon SM8850, Android 16, SDK 36; 14.88 GiB RAM; eight ARM64 cores in a 6+2 arrangement, six at 3.63 GHz and two at 4.74 GHz). A saved on-device library listing shows an OEM HTP v81 runtime (`libQnnHtpV81`) that this deployment does not use: the execution backend is the official Android ARM64 Q4 CPU runner.

The demo is complete on the handset: native microphone capture, streaming text, and generated speech pass end to end in the embedded UI, and the application cold-launches and serves a fresh microphone inference with no computer, cable, or network attached. It is a CPU path — the NPU port is §9 — and it supplies the first measurement of this model resident on real mobile hardware, the quantity §6 could not derive from static files.

Two structural properties bound the numbers: complete audio files are submitted before inference begins (batch timings after utterance completion, excluding capture and VAD), and reported p95 values at $n \leq 10$ are sample maxima.

8.2 Residency: the static-to-resident gap

Table 8 is the central result. The Q4 bundle occupies 1.001 GiB as static files (§6); the model process resident set peaks at 1873.4 MiB, or 1.829 GiB — a factor of 1.83. Static size underpredicts residency by nearly half, which is why the ledger of §6 is a lower bound and not a budget.

The kernel high-water mark (`VmHWM`) is 1874.2 MiB, about 0.8 MiB above the sampled peak, so 0.25 s sampling slightly understates the true maximum.

Attribution. A follow-up batch with a retained runner load log and `smaps` capture names the gap. At the original 4096 context, active residency splits into 697.6 MiB of GGUF-file-backed pages against 1041.2 MiB of anonymous memory, and the load log declares the major anonymous allocations (Table 7): the largest single item is a 555.75 MiB CPU weight *repack* — a second, kernel-friendly copy of the backbone weights that coexists with their 652.3 MiB mmapped originals — followed by three fixed compute buffers totaling 458.7 MiB. The formerly unexplained ∼800 MiB is therefore repacked weights and compute geometry, not hidden model files.

Context length is a weak lever and audio length a strong one: shrinking context 4096 to 512 saves only 42.9 MiB, while moving from the 4.9 s to the 18.4 s input raises active residency by roughly 125 MiB at every context size (1870.6 MiB at context 4096 — consistent with the original run's 1873.4 MiB peak, which the long-ASR workload set). The longest 29.4 s diagnostic utterance drives `VmHWM` to 1919.3 MiB with repacking enabled.

The joint verdict. Disabling repacking (`-no-repack`, context 512) removes 538 MiB of active residency and preserves the matched 18-utterance diagnostic within the 0.3 pp rule on every split — but peaks at 1365.3 MiB `VmHWM` on the 29.4 s utterance and roughly doubles to triples request latency (single ordered runs; first audio exceeded 1.2 s). Configurations pushed to 1302.9–1320.2 MiB via smaller micro-batches and KV quantization all fail the competing-speech split by $+0.48$ to $+1.90$ pp. Stating the gate in explicit units — $\leq$1331 MiB (1.3 GiB); the stricter decimal reading is 1240 MiB — no tested CPU configuration passes memory, quality, and latency simultaneously. The CPU Q4 deployment is a working demo and research baseline; it is not an accepted product configuration.

8.3 Latency

Table 9 reports the three workloads. Cold start (process restart, tap to ready) was 823.6/899.3/906.2 ms over three launches; model pages remained in the Linux page cache, so this is not a post-reboot cold load.

The conversational workload reaches first audio at a median 799.7 ms after file submission. The gate of §11 asks for first audible PCM within 500 ms of VAD close, and no VAD exists in this pipeline, so the two quantities are not the same measurement and 799.7 ms should not be read as a gate result.

Table 6: Local Q4 inference-technique matrix on Apple M2 Metal. Values are median/p95 over five runs per sample and configuration. CLI model time excludes model load; wall time in the public data includes process startup and file-cache effects.

KV	Flash	Sample	Exact	KV MiB	Encode ms *med/p95*	Gen tok/s *med/p95*	Model ms *med/p95*
FP16	off	asr	5/5	48.0	449.0 / 629.8	24.3 / 25.6	3425.8 / 3555.5
FP16	off	question	5/5	48.0	140.0 / 298.4	29.1 / 57.4	816.9 / 1590.3
FP16	on	asr	5/5	48.0	439.0 / 592.4	23.2 / 26.4	3389.3 / 3855.5
FP16	on	question	5/5	48.0	172.0 / 196.8	28.6 / 38.8	891.4 / 1096.6
Q4_0	on	asr	5/5	13.5	413.0 / 476.8	25.7 / 26.1	3122.0 / 3285.7
Q4_0	on	question	5/5	13.5	139.0 / 148.8	32.6 / 48.6	738.1 / 904.6
Q8_0	on	asr	5/5	25.5	486.0 / 780.4	26.5 / 28.1	3254.8 / 3780.5
Q8_0	on	question	5/5	25.5	137.0 / 195.4	41.7 / 67.7	627.5 / 982.5

Table 7: Load-log declared allocations behind the anonymous residency (context 4096). Declared buffers need not be simultaneously resident, so the column does not sum to `smaps` RSS; log and `smaps` agree on the attribution.

Runtime allocation	Declared size
Main-model CPU repack	555.75 MiB
Audio-encoder compute buffer	195.19 MiB
Main compute buffer	132.00 MiB
Detokenizer compute buffer	131.50 MiB
Main KV, context 4096, FP16	48.00 MiB
Detokenizer CPU repack	19.27 MiB
Detokenizer sliding-window KV	2.25 MiB

Table 8: Peak resident memory on the phone, from `/proc/<pid>/status` sampled at a 0.25 s target interval (172 samples). `VmRSS` unless noted. The 1.3 GB gate is exceeded by the model process alone.

Process	Peak (MiB)	Peak (bytes)
Model (runner)	1873.4	1,964,384,256
kernel `VmHWM`	1874.2	1,965,240,320
App (UI/service)	297.0	311,398,400
Combined (sampled)	2169.1	2,274,426,880
Static Q4 bundle	*1025.0*	*1,075,000,000*

8.4 CPU and thermal

A separate probe attributes 30.62–33.49 core-seconds to the model process over 3.93–4.26 s of wall time: an average of 7.86 cores in flight during generation. The UI/service process rounds to 0.00 s at the kernel's 100 Hz accounting resolution, which reflects rounding rather than idleness; inference compute lives in the model process. The probe reused a live persistent session and is therefore not comparable to the fixed-output chat benchmark above.

That the model saturates roughly eight cores is itself the strongest available evidence that no accelerator is in play: the compute is accounted for on the CPU.

Thermals moved little across a 52 s window: CPU maximum 59.0 to 59.4 °C, GPU 47.8 to 52.9 °C, NSP 50.9 to 50.5 °C, skin 28.54 to 35.44 °C. These are two snapshots, not a time series, and 52 s is not a sustained-load test. The phone was charging throughout, so battery drain was recorded but is not a valid power measurement and is not reported.

Table 9: Phone latency, median/p95 in ms, warm-ups excluded. p95 at these sample sizes is the maximum observation. ASR workloads emit no audio.

Workload	n	First text	First audio	Total
ASR (short)	10	250.9 / 272.4	—	442.5 / 470.9
ASR (long)	5	854.8 / 856.0	—	1554.4 / 1583.5
Chat	5	677.2 / 700.5	799.7 / 837.4	6786.5 / 6876.9

8.5 Transcript integrity

All 15 scored ASR runs returned exact normalized transcripts (0.00% WER) against the two official demo samples, and all five chat runs produced byte-identical output (1,096,320 bytes of float32 PCM, 11.42 s at 24 kHz) at temperature 0.

This is the same two-sample check as the Q4 matrix of §7, extended from Apple M2 Metal to Android CPU: 40/40 exact there, 15/15 exact here, same bundle.

The follow-up batch additionally ported a *matched* 18-utterance diagnostic to the device — same corpus and deterministic 10 dB Gaussian and 5 dB competing-speech variants as §5, 324 requests with zero failures. Phone Q4 scores 3.57 / 7.14 / 54.29 % (repack) and 3.57 / 6.90 / 54.29 % (no-repack) against the L4 BF16 reference's 2.86 / 6.43 / 55.71 %. Every gap is within one to two words: at n=18 (≈140 reference words) a single substituted word moves WER by roughly 0.7 pp, so the instrument is coarser than the 0.3 pp gate it is asked to decide. The corpus must be scaled (to ≥100 utterances) before any quantized-WER adjudication.

Byte-count and duration integrity were verified; NaN/Inf validation and on-device audio-text retranscription were not run.

9 Toward an NPU port

9.1 Toolchain gate

The NPU path required establishing that the phone's Hexagon generation is reachable at all. It is, twice over: Qualcomm AI Hub exposes the exact silicon generation — a Snapdragon 8 Elite Gen 5 QRD (chipset `sm8850`, HTP v81) plus the Galaxy S26 family — and the public QAIRT 2.48.0 community archive ships the complete v81 target payload for Linux and Windows hosts. The phone itself carries an OEM QAIRT 2.37.0 v81 runtime. macOS is not a supported QAIRT host, so local compilation requires an ARM64 Linux environment; AI Hub covers the interim. Results below come from the QRD, a reference device of the phone's silicon generation.

9.2 FP16 transfer to HTP v81

All eight fixed-shape graphs were re-submitted, unmodified, as strict FP16 context binaries for v81, and every one places fully on the NPU at 1.46–1.88× lower component latency than v73 (Table 10). The depth/RQ decoder keeps exact tokens after one boundary correction (`-truncate_64bit_io`; int32 at the QNN boundary, exactly equal to the int64 goldens). Peaks sit at 117.5–118.0 MiB, a near-uniform +25–31 MiB over v73 regardless of component size — consistent with a runtime-baseline change, not per-model growth.

The stable result matters most: the numerical classes are identical across the silicon generation and the toolchain revision. Backbone and depth pass on both; the encoder reproduces v73's accepted signature (cosine 0.998423 versus 0.998424); both detokenizer probes remain unusable on both (v81 angle cosine 0.26–0.28). The partition of Figure 1 is a property of the model, not of v73 or an old compiler — and FP32 host detokenization is now a design decision validated across a hardware generation.

9.3 Quantization smoke (W8A16)

A first quantization pass validated mechanics, not quality. The depth/RQ decoder — chosen because its acceptance is binary — was blocked before compilation: seven `to_logits` weights are tied, consumed by both a `Gemm` (output projection) and a `Gather` (embedding lookup), and the quantizer requires each initializer duplicated per consumer. This is a mechanical graph-preparation fix, with one watch item: duplication quantizes seven large tensors twice, so the resulting binary size needs checking.

The conv-prefill fallback completed the full pipeline — quantize, strict v81 compile, profile, inference — with full NPU placement, a 64.6 MiB binary (49.6% of FP16), and 0.977 ms latency (1.88× faster than its current-toolchain FP16 comparator). Its output sits outside the FP16 transfer tolerance (max abs error 0.0230, cosine 0.9999955) — a number that is currently uninformative in either direction: calibration used a single example (the 80-sample real-embedding set is captured and awaits export approval), and quantized graphs are accepted on transcripts and tokens (§5), not tensor distance. The calibrated downstream run is queued and will decide it.

9.4 Per-step arithmetic and the next gate

A deliberately naive floor: assuming the LFM2 backbone layout of ten convolution and six attention blocks (six is fixed by the KV-cache accounting of §6), one generation step on v81 costs $10 \times 1.831 + 6 \times 1.673 + 16.778 \approx 45$ ms of pure component time — about 22 steps/s, with the depth decoder alone taking 37% of the budget. The no-summing rule of Figure 2 applies in full: this arithmetic excludes every transfer, graph switch, and cache-management cost, which on an explicit-cache-I/O design is exactly where the budget is most likely to go. It motivates, and does not replace, an on-device step-rate microbenchmark with QNN registered shared buffers as the next gate before any application work.

10 Comparators and candidate selection

10.1 Qualcomm speech component cards

Table 11 reports full-NPU proxy scorecards from `qai-hub-models` 0.57.3 for the alternative pipeline components. These are fixed component invocations, not end-to-end pipeline latency.

Whisper Small W8A16 is instructive: quantization makes the encoder 2.89× slower while making the decoder roughly 35% faster. Quantization must therefore be profiled stage by stage; a single whole-model figure would have hidden both effects.

10.2 Decision

Table 12 records the candidate outcome. LFM2.5-Audio continues as the integrated research baseline under the gates of §11. Moonshine Base → Qwen3-0.6B → Pocket TTS [Useful Sensors, 2024, Qwen Team, 2025, Kyutai Labs, 2025] is the recommended next challenger: it creates the most weight-footprint headroom while keeping the text model on a known Qualcomm path. It still needs export, quality, and host-TTS measurements before it can replace LFM.

LLaMA-Omni2-0.5B [Fang et al., 2025] is excluded on evidence rather than principle: the size label refers to the language backbone, while the checkpoint alone is 3.857 GB in BF16.

11 Gates and next steps

11.1 Demo acceptance gates

Table 13 lists provisional thresholds for a demo, alongside what the CPU phone run of §8 does and does not say about each. The thresholds are provisional in a specific sense: the memory gate is a guess until app-available RAM on the real device is confirmed, and should be revised the moment it is.

Table 10: FP16 transfer matrix, v73 (QCS8550 proxy) versus v81 (8 Elite Gen 5 QRD). Placement is identical and fully on-NPU on both generations. Caveat: v73 numbers were compiled by the AI Hub toolchain current at their submission date; v81 numbers use the current toolchain. Differences reflect silicon generation and toolchain version jointly; neither is isolated. †Outside the strict 10^{-3} tensor gate; the identical v73 feature signature passed the downstream acceptance test (§5); the downstream check on v81 features is queued.

Component	Layers	v73	v81	Speedup	Peak v81	Numerics (both gens)
FastConformer + adapter	1215	4.786 ms	3.185 ms	1.50×	117.6 MiB	task pass†
Backbone conv prefill	36	2.693 ms	1.833 ms	1.47×	117.7 MiB	pass
Backbone attention prefill	62	2.444 ms	1.667 ms	1.47×	117.7 MiB	pass
Backbone conv cached decode	32	2.695 ms	1.831 ms	1.47×	117.5 MiB	pass
Backbone attention cached decode	61	2.452 ms	1.673 ms	1.47×	117.8 MiB	pass
Depth/RQ decoder	2953	25.540 ms	16.778 ms	1.52×	117.8 MiB	tokens exact; embedding pass
Detokenizer T=4	371	1.880 ms	1.284 ms	1.46×	117.9 MiB	mismatch
Detokenizer T=8	371	2.577 ms	1.372 ms	1.88×	118.0 MiB	mismatch

Table 11: Qualcomm proxy component cards for comparator models (`qai-hub-models` 0.57.3). Fixed component invocations, not end-to-end pipeline latency.

Model	Encoder/main	Recurrent	Use
Zipformer	8.895 ms / 0.71 s chunk	0.075 ms dec.; 0.187 ms joiner	Streaming fallback
Whisper Tiny	25.307 ms	2.459 ms dec.	Small control
Whisper Base	47.978 ms	4.202 ms dec.	Middle control
Whisper Small FP16	130.318 ms	12.074 ms dec.	Overlap-robust control
Whisper Small W8A16	376.895 ms	7.856 ms dec.	Memory tradeoff
Piper TTS	30.344 ms enc.; 15.189 ms flow	3.018 ms dec.	Audio fallback

Table 12: Candidate decisions. Footprint figures come from different sources and are not directly comparable.

Candidate	Footprint view	Decision
LFM2.5-Audio	Q4 bundle 1.001 GiB locally	Continue with gates
Moonshine + Qwen3-0.6B + Pocket	Est. 0.55–0.66 GB weights	Profile next
Zipformer + Qwen3-0.6B + Piper	~0.84 GB from component cards	Demo insurance
Mini-Omni [Xie & Wu, 2024]	~976M total est.	Later control
LLaMA-Omni2-0.5B	Checkpoint alone 3.857 GB BF16	Exclude

The memory gate is now stated in explicit units — ≤1331 MiB (1.3 GiB), with the stricter decimal reading 1.3 GB = 1240 MiB noted for reference — and it is missed: the best quality-preserving configuration peaks at 1365.3 MiB, and every configuration under the threshold fails the competing-speech quality split (§8). The frontier is three-way: memory, quality, and latency trade against each other and no tested CPU configuration holds all three. Three of the remaining gates are not merely unmeasured but *not exercisable* on a CPU deployment: there are no NPU shards for a placement gate to inspect, no VAD for a responsiveness gate to trigger from, and no multi-turn workload for a retention gate to score. This is a property of the deployment, not an oversight in it.

11.2 Next steps

1. Unblock depth quantization: duplicate the seven tied `to_logits` initializers, calibrate with the captured 80-sample real-embedding set, and accept on exact tokens — then W4A16 the backbone, since W8A16 alone roughly doubles the Q4 weight footprint and exceeds the memory gate on weights.
2. Run the on-device step-rate microbenchmark with QNN registered shared buffers: prefill plus N cached decode steps plus depth, measuring transfer overhead against the ∼45 ms naive floor of §9, before any application integration.
3. Recover repack memory without repack latency on the CPU baseline: selective repacking, releasing mmapped originals of repacked tensors, or staged encoder/detokenizer loading (the encoder's fixed 195 MiB buffer and length-scaled activations are the peak drivers).
4. Scale the matched diagnostic to ≥100 utterances so the 0.3 pp quality gate becomes decidable; the current n=18 resolution is ∼0.7 pp per word.
5. Measure the official FP32 detokenizer on the intended host path; the FP16 HTP route is closed on both tested silicon generations.
6. Run the unplugged 30-minute power and thermal series

Table 13: Provisional demo acceptance gates and their status after the CPU phone run (§8). The phone is not the glasses and not an NPU path; "not exercisable" means the deployment contains nothing the gate could measure.

Gate	Provisional threshold	Phone (CPU) status
Memory	Model-process peak $\leq$ 1331 MiB (1.3 GiB)	**Missed**: best quality-preserving config 1365.3 MiB
Placement	No unsupported main-backbone op; no undeclared CPU fallback	Not exercisable: no NPU shards in this path
ASR quality	Quantized WER increase $\leq$ 0.3 pp per split	Screened: matched n=18 set; resolution ($\sim$0.7 pp/word) coarser than the gate
Responsiveness	First audible PCM $\leq$ 500 ms after VAD close	Not exercisable: no VAD; 799.7 ms post-submission
Context	Two-turn retention within five points of BF16 golden	Not exercisable: all workloads single-turn
Audio	No NaN/Inf, truncation, or audio-text regression	Partial: byte/duration integrity only

(the batch's cached 99.6 °C CPU readings after 324 requests are warning evidence, not a sustained result).

7. Profile the modular challenger in parallel — Qwen3-0.6B, then Moonshine Base, then Pocket TTS — with Zipformer and Piper as swaps.
8. Confirm the physical glasses SoC, app-available RAM, QNN/Voice AI SDK version, and deployment interface; repeat the accepted configuration there.

12 Limitations

The proxy is not the target. QCS8550 latency is not AR1 latency, and no AR1/AR1+/5100 device was available for any measurement in this report.

Component latency and memory cannot be summed into an end-to-end figure. The components have different shapes, invocation counts, compiled-context overheads, and transfer boundaries, and Figure 2 should not be read as a budget.

Backbone evidence is representative rather than one full compiled network. The ASR diagnostic uses 18 utterances and synthetic noise, and is not a glasses microphone benchmark. Host FP32 detokenizer viability is proposed, not measured on target. The Q4 matrix uses two official samples on Apple M2 Metal and establishes neither general quality, nor Qualcomm placement, nor glasses memory.

The phone deployment is a CPU path on a flagship handset. It is not an NPU result, not AR1, and not the glasses, and its residency figure transfers to the glasses only as a lower bound on what a Q4 CPU runner costs. Device identity is now on record (14.88 GiB RAM, ARM64, 6+2 cores), and a saved library listing shows an unused OEM HTP v81 runtime, so the CPU-only claim rests on positive evidence. Phone thermals remain snapshots — the batch's post-run cached 99.6 °C CPU readings are warning evidence, not a sustained-load profile — and the phone was charging, so no power figure is reported. Phone p95 values at $n \leq 10$ are sample maxima; resident-memory peaks are sampled at 0.25 s; and the no-repack latency comparison is single ordered runs, not a distribution.

The v81 results come from a cloud reference device of the phone's silicon generation, profiled through AI Hub: isolated fixed-shape component calls with no transfers, cache management, or end-to-end path, on a QRD rather than retail hardware, and with silicon and toolchain versions jointly confounded against the v73 baseline. The W8A16 smoke is a mechanics result only: its calibration was a single example and its tensor-tolerance verdict is not the prescribed acceptance test for quantized graphs, so no quantization-quality conclusion is drawn from it.

Finally, licensing is not settled: the LFM Open License v1.0 contains a commercial-use revenue threshold, and the candidate components carry separate attribution and license terms.

References

Liquid AI. LFM2.5-Audio-1.5B model card. https://huggingface.co/LiquidAI/LFM2.5-Audio-1.5B.

Liquid AI. liquid-audio source repository. https://github.com/Liquid4All/liquid-audio.

Qualcomm. AI Hub model catalog. https://aihub.qualcomm.com/models/.

Qualcomm. Zipformer model card. https://aihub.qualcomm.com/models/zipformer.

Qualcomm. Whisper Small model card. https://aihub.qualcomm.com/models/whisper_small.

Qualcomm. PiperTTS English model card. https://aihub.qualcomm.com/models/pipertts_en.

Useful Sensors. Moonshine Base model card. https://huggingface.co/UsefulSensors/moonshine-base. arXiv:2410.15608.

Qwen Team. Qwen3-0.6B model card. https://huggingface.co/Qwen/Qwen3-0.6B. Qualcomm card: https://aihub.qualcomm.com/models/qwen3_0_6b.

Kyutai Labs. Pocket TTS. https://github.com/kyutai-labs/pocket-tts.

Z. Xie and C. Wu. Mini-Omni: Language models can hear, talk while thinking in streaming. arXiv:2408.16725, 2024.

Q. Fang et al. LLaMA-Omni2: LLM-based real-time spoken chatbot with autoregressive streaming speech synthesis. arXiv:2505.02625, 2025.

A Reproducibility

Phone-run provenance. The published `REPORT.md` is the profiler's generated output plus a hand-inserted CPU-probe summary section sourced from `cpu_probe.json`; `profile.json` was likewise augmented with that probe summary, which the profiler does not itself emit. Both edits are numerically faithful: every published value was verified

Table 14: Pinned revisions and environments. Hashes are unbroken for copy-paste.

Item	Pinned value
LFM model revision	`c362a0625dfe45aa588dce5f0ada28a7e5707628`
Q4 GGUF / runner	`7d525f883a077e20afb782f2ff618edcae0e39e4`; build 7641 (`68d8edf2`); Apple M2 Metal
Liquid Audio source	`19e65845923a7f136442c95137884ec61eb386aa`
Qualcomm proxy	QCS8550 (Proxy), Android 12, Hexagon v73, QNN HTP FP16
Comparator catalog	`qai-hub-models` 0.57.3; QAIRT 2.45.0 scorecards
Full-model golden	PyTorch 2.8.0, Transformers 4.56.1, NVIDIA L4 BF16; seed 20260715
Phone target	Nubia NX809J, Snapdragon SM8850, Android 16 (SDK 36); official Android ARM64 Q4 CPU runner
Phone run	`20260716t120118z`; 172 resource samples at 0.25 s target interval; 100 Hz CPU accounting
Phone batch	`experiment_batch_20260717`; attribution, sweeps, and matched diagnostic (324 requests)
NPU toolchain	AI Hub target Snapdragon 8 Elite Gen 5 QRD (`sm8850`, HTP v81); `qai-hub` 0.52.0; public QAIRT 2.48.0; phone OEM QNN runtime 2.37.0
Depth calibration	80 embeddings, audio frames 0–79, generation steps 6–120 of the pinned turn NPZ SHA-256 `2e3e4e282385339c1f09574107a64cde31457b40c80c55ea064a2911bb8b35e2`

by regenerating the report from the raw JSON and diffing, and the `runs.csv` aggregates recompute to the published medians and p95 values exactly. The run directory was imported into the repository on the following day, so file modification times postdate the run stamp. The profiler's `lfm-server.log` was excluded from the published directory; the directory is otherwise what the script produces. The v81 transfer and W8A16 reports were likewise handwritten from script-generated raw archives with all AI Hub job identifiers retained, and later runs keep their server logs. Archived JSON evidence had absolute local path strings redacted post-capture; measured values are unchanged.